Md Ahsanul Haque, Ph.D.
MAT-120

Midterm

Due Date: April 5, 2026

Name

Student ID

Section

DISTRIBUTION OF MARKS		
Question No.	**Full Marks**	**Obtained Marks**
1		
2		
3		
4		
5		
6		
Total		

Instructions:

- Write your full name (first **and** last name) **inside** the provided box.
- Clearly write your student ID and section number.
- This assignment contains 6 problems. It is **YOUR responsibility** to check that your copy has all 6 problems.
- Answer **5** questions. If you answer 6, the first 5 will be graded.
- Make sure to read the instructions for each problem carefully.
- Use correct mathematical notation.
- Write cohesively and follow a logical sequence of steps.
- Check that your answer can be read and understood by the grader.

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1. (10 points)

   Differentiate the following functions:

   (a) $y = \cos(e^{2x} + 5x)$

   (b) $y = x^4 \ln(3x^2 + 1)$

   (c) $y = \left(\dfrac{2x+3}{x-1}\right)^4$

   (d) $y = e^{\tan x} + \sec(x^2)$

Extra Space

2. (10 points) Consider the function:

$$f(x) = \begin{cases} \dfrac{3x^2+2}{x^2-1} & \text{if } x < -2 \\ 4 & \text{if } x = -2 \\ x^2 - 1 & \text{if } x > -2 \end{cases}$$

(a) Find $\lim_{x \to -2^-} f(x)$ and $\lim_{x \to -2^+} f(x)$

(b) Does $\lim_{x \to -2} f(x)$ exist? Justify your answer.

(c) Find $\lim_{x \to -\infty} f(x)$ and $\lim_{x \to \infty} f(x)$

(d) Identify all points of discontinuity and classify them.

Extra Space

3. (10 points)

   (a) Let $f(x) = \dfrac{\frac{1}{x+2} - \frac{1}{2}}{x}$.
   Find $\lim_{x\to 0} f(x)$ using algebraic manipulation.

   (b) Determine if $g(x) = \dfrac{x^2 - 4x + 3}{x - 1}$ is continuous at $x = 1$. If not, is the discontinuity removable?

   (c) Find the value of c that makes $h(x) = \begin{cases} cx + 3 & \text{if } x \leq 3 \\ cx^2 - 3 & \text{if } x > 3 \end{cases}$ continuous everywhere.

   (d) Investigate the solution of $x^3 + x^2 - 2x = 1$ in the interval $[-1, 1]$.

Extra Space

4. (10 points) Analyze the following limits:
   (a) $\lim_{x\to 0} \dfrac{\tan(4x)}{7x}$
   (b) $\lim_{x\to 4} \dfrac{\sqrt{x}-2}{x-4}$
   (c) $\lim_{x\to -\infty} \frac{\sqrt{9x^2+1}}{2x-5}$
   (d) $\lim_{x\to 0} x^2 \cos\left(\frac{1}{x^2}\right)$ using the Squeeze Theorem.

Extra Space

5. (10 points) Let $f(x) = x^3 - 12x$ and $g(x) = \frac{1}{\sqrt{x}}$
   (a) Find the equation of the tangent line of f at $x = 1$.
   (b) Find the average rate of change of f on $[1, 4]$ and compare it with the instantaneous rate at $x = 2.5$.
   (c) Use the **limit definition** of the derivative to find $g'(x)$. Show the rationalization step clearly.
   (d) Find the equation of the normal line (the line perpendicular to the tangent line) of g at $x = 4$.

Extra Space

6. (10 points) Consider the curve defined by

$$x^2 + y^2 + xy = 7.$$

   (a) Find $\dfrac{dy}{dx}$ using implicit differentiation.

   (b) Find all points where the tangent line is horizontal.

   (c) Find all points where the tangent line is vertical.

   (d) Find $\dfrac{d^2y}{dx^2}$ in terms of x and y.

Extra Space